National Institute of Technology Rourkela
Department of Mathematics
MA4271 Scientific Computing Laboratory, Assignment-1, 2

Instructions: Write your Name and Roll no. on the answer script. Rename the files with your Roll No. Copy your program code (.m file if using MATLAB) and output to the answer script. Record/Submit your answer in one (.pdf format) in the same place.

1. Perform the Bisection method upto 50 iteration and find the value of $\sqrt{2}$. Also write your output in the table given below

Iteration	a	b	$c = \frac{a+b}{2}$	$f(a)$	$f(b)$	$f(c)$
0						
1						
2						
$\vdots$						
50						

2. Use the bisection method to find the root of $\cos x = xe^x$, accurate to at least 7 significant digits. (Take the problem of your choice.) Use the following stopping criterion.

   (a) Absolute error

   (b) Relative error

   (c) Functional value

	a	b	Approximate value of root	No of Iteration required
Absolute Error				
Relative Error				
Functional value				

3. Take the initial guess x_0 as a mid point of $[a, b]$ and repeat the question no. 2 for N-R method first and then the fixed point iteration method.

	a	b	x_0	Approximate value of root	No. of Iteration required
Absolute Error					
Relative Error					
Functional value					

   Write your comments/observations regarding the above methods/results.

4. In assignment-2, repeat the above question 2 with the secant and regula–falsi methods. Finally print the results in tabular forms.

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